It is obvious that if the expected value ($\hat{y}_i$) is close to the observed value (y_i), the residual will be small. Since the magnitude of the residual is determined by the values of '*a*' and *'b'*, estimates of these coefficients are obtained by minimizing the sum of the squared residuals, *E(a,b).*

Differentiation of *E(a,b)* with respect to *'a'* and *'b'* and equating them to zero constitute a set of two equations as described below:

$$\frac{\partial E(a,b)}{\partial a} = -2\sum_{i=1}^{n}(y_i - a - bx_i) = 0$$

$$\frac{\partial E(a,b)}{\partial b} = -2\sum_{i=1}^{n}x_i(y_i - a - bx_i) = 0$$

These give

$$na + b\sum_{i=1}^{n}x_i = \sum_{i=1}^{n}y_i$$

$$a\sum_{i=1}^{n}x_i + b\sum_{i=1}^{n}x_i^2 = \sum_{i=1}^{n}x_iy_i$$

These equations are popularly known as **normal equations.** Solving these equations for '*a*' and *'b'* yield the estimates $\hat{a}$ and $\hat{b}$.

$$\hat{a} = \overline{y} - \hat{b}\overline{x}$$

and

$$\hat{b} = \frac{\frac{1}{n}\sum_{i=1}^{n}x_iy_i - \overline{x}\,\overline{y}}{\frac{1}{n}\sum_{i=1}^{n}x_i^2 - \overline{x}^2}$$

It may be seen that in the estimate of '*b*', the numerator and denominator are respectively the sample covariance between *X* and *Y,* and the sample variance of *X*. Hence, the estimate of *'b'* may be expressed as

$$\hat{b} = \frac{Cov(X,Y)}{V(X)}$$

Further, it may be noted that for notational convenience the denominator of $\hat{b}$ above is mentioned as variance of *X*. But, the definition of sample variance remains valid as defined in Chapter I, that is, $\frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \overline{x}^2\right)$.

From Chapter 4, the above estimate can be expressed using, r_{XY}, Pearson's coefficient of the simple correlation between *X* and *Y,* as

$$\hat{b} = r_{XY}\frac{SD(Y)}{SD(X)}.$$

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